

A NOTE OF TOPOLOGICAL METHODS IN GROUP THEORY

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ABSTRACT. This is a course note of 2026 Spring MATH 569, taught by Daniel Groves at UIC, on the topic of Topological Methods in Group Theory. Major reference textbooks are: [Hat01], [Bro82], [Geo08].

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1. ASPHERICAL SPACES AND SAMPLE EARLY THEOREMS

In this course we will explore the relationship between aspherical spaces and (discrete) groups. A convention: unless otherwise stated, all spaces in this course will be CW complexes.

Definition 1.1 ($K(G, 1)$). Let G be a group. A $K(G, 1)$ is a connected space such that $\pi_1(K(G, 1)) \cong G$ and whose universal covering space $\widehat{K}(G, 1)$ is contractible.

It is also called *the classifying space* or *Eilenberg-MacLane space*.

Definition 1.2 (Aspherical Space). If X is a connected space such that $\widetilde{X}$ is contractible, then we say X is a $K(\pi, 1)$ (i.e. $K(\pi_1(X), 1)$). We call such X *aspherical*.

Theorem 1.3 ([Hat01, Example 1.B.7, Theorem 1.B.8, Proposition 1.B.9]). *For any group G , there exists a (CW) $K(G, 1)$ which is unique up to homotopy equivalence. In fact, for any two groups G, H , there exists a bijection*

$$\mathrm{Hom}(G, H) \leftrightarrow \{\text{based maps } K(G, 1) \rightarrow K(H, 1)\} / \text{based homotopy}$$

Let X be a space and x_0 a point in X . Recall that $\pi_0(X)$ represents the path components of space X . For $n \geq 1$, $\pi_n(X, x_0)$ is the group of based homotopy classes of maps $(\mathbb{S}^n, \star) \rightarrow (X, x_0)$.

For $n \geq 2$, any map $(\mathbb{S}^n, \star) \rightarrow (X, x_0)$ lifts to $(\mathbb{S}^n, \star) \rightarrow (\widetilde{X}, \widetilde{x}_0)$ (Since $\pi_1(\mathbb{S}^n, \star)$ is trivial). If X is aspherical, then $(\mathbb{S}^n, \star) \rightarrow (\widetilde{X}, \widetilde{x}_0)$ is null-homotopic. Hence $\pi_n(X, x_0) = 0$. This proves one direction of Theorem 1.4. The other direction is harder, and we leave it for future discussion.

Theorem 1.4 (Whitehead's Theorem, [Hat01, Theorem 4.5]). *A CW complex X is aspherical if and only if it's connected and for all $n \geq 2$, $\pi_n(X, x_0) = 0$.*

Example 1.5 (Aspherical Spaces). (1) A point.

(2) $\mathbb{S}^1$.

(3) Products of aspherical spaces (corresponding to direct products of groups).

(4) Wedges of aspherical spaces (corresponding to free products of groups).

(5) Graphs (1-dimensional CW complexes).

(6) Closed orientable surfaces with genus ≥ 1 .

(7) Complete Riemannian manifolds of non-positive sectional curvature (Cartan-Hadamard theorem).

(8) Complete locally CAT(0) spaces (A variant of Cartan-Hadamard theorem. See for example [BH99, Theorem II.4.1]).

Example 1.6 (Non-aspherical Spaces). Connect sums of aspherical spaces typically produce non-aspherical spaces.

Question 1.7. • What kind of $K(G, 1)$ spaces does [insert any interesting group] have?

- For [insert any interesting group], which $K(G, 1)$ is “better”?
- What do properties of $K(G, 1)$ say about G ?

Below we list some sample early theorems without proof.

Theorem 1.8. *A group G is finitely generated if and only if G has a $K(G, 1)$ with a finite 1-skeleton.*

Theorem 1.9. *A group G is finitely presented if and only if G has a $K(G, 1)$ with a finite 2-skeleton.*

Theorem 1.10. *If G has non-trivial torsion, then there does not exist any finite-dimensional $K(G, 1)$.*

Another way to phrase Theorem 1.10 is: if G has a finite-dimensional $K(G, 1)$, then it's torsion free.

Observe that those theorems all related to some sort of finiteness of $K(G, 1)$. We introduce some notions of finiteness below.

Definition 1.11. A group G is of *type F_n* if it has a $K(G, 1)$ with finite n -skeleton.

A group G is of *type F_∞* if it's of type F_n for all n .

A group G is of *type F* if it has a finite $K(G, 1)$.

A group G has *finite geometric dimension* if it has a finite-dimensional $K(G, 1)$. The *geometric dimension*, denoted as $\text{gd}(G)$, is the minimal dimension of all of its $K(G, 1)$.

It is sometimes difficult to understand the $K(G, 1)$ of a group. We thus use multiple invariance to help determine those types. If G is a group and X is its $K(G, 1)$, then the algebro-topological invariance of X is invariant of G . In particular, we look at the cohomology of groups (with trivial coefficients):

$$H^*(G, \mathbb{Z}) := H^*(X, \mathbb{Z})$$

2. COHOMOLOGY OF GROUPS

Definition 2.1. We say a space (Y, y_0) is *n -connected* if $\pi_i(Y, y_0) = 0$ for all $0 \leq i \leq n$.

Other than the trivial coefficient $\mathbb{Z}$, we use also group rings to calculate the cohomology of groups to get different properties.

Definition 2.2. Let G be a group. The group ring $\mathbb{Z}G$ is the set of finite $\mathbb{Z}$ -linear combinations of elements of G .

Remark 2.3. We may also use $\mathbb{R}$, fields, or PIDs, etc. instead of $\mathbb{Z}$ to get different group rings.

Example 2.4. Let $G = \mathbb{Z} = \langle t \rangle$. Then $\mathbb{Z}G = \mathbb{Z}[t, t^{-1}]$.

Example 2.5. Let $G = \mathbb{Z}/n\mathbb{Z} = \langle t \rangle$. Then $\mathbb{Z}G = \mathbb{Z}[t]/(t^n - 1)$.

Notice that $g^n = 1$ for some non-trivial element $g \in G$, i.e. G has a non-trivial torsion, then we have:

$$0 = g^n - 1 = (g - 1)(g^{n-1} + \cdots + g + 1)$$

which implies that $g - 1$ is a non-zero zero-divisor. The other direction of this fact is an open question.

Question 2.6 (Kaplansky's Conjecture). If G is torsion-free, then $\mathbb{Z}G$ has no non-zero zero-divisor.

A motivation for Question 2.6 is that we may embed $\mathbb{Z}G$ into its division ring, so we wish to have no non-zero zero-divisors in $\mathbb{Z}G$.

Typically G is not abelian, so $\mathbb{Z}G$ is not a commutative ring. Left and right $\mathbb{Z}G$ -modules are typically different. By default, we use left $\mathbb{Z}G$ -modules.

Remark 2.7. If M is a left $\mathbb{Z}G$ -module, it can be turned into a right $\mathbb{Z}G$ -module by defining a right G -action $mg = g^{-1}m$.

Definition 2.8 (Free resolution). A free resolution of $\mathbb{Z}$ (the trivial $\mathbb{Z}G$ -module) by free $\mathbb{Z}G$ -module is a resolution

$$\cdots \rightarrow F_1 \rightarrow F_0 \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0$$

that is exact, and every F_i is a free $\mathbb{Z}G$ -module (i.e. $(\mathbb{Z}G)^n$ for some n).

Sometimes we also use resolutions by projective $\mathbb{Z}G$ -modules (See Definition 5.5).

Remark 2.9. The free resolution arises naturally from $K(G, 1)$ in the following manner. Say X is a CW complex $K(G, 1)$ (recall Definition 1.1), then any i -cell in $\tilde{X}$ is $g\sigma$ for some $g \in G$ and some i -cell σ in X . So we can view the cellular i -chain $C_i(\tilde{X}; \mathbb{Z})$ as a free $\mathbb{Z}G$ -module whose basis is in bijective with i -cells in X .

Since $\tilde{X}$ is contractible, $H_*(\tilde{X}) = 0$. That is, the complex

$$\cdots \rightarrow C_{n+1}(\tilde{X}) \xrightarrow{\partial_{n+1}} C_n \xrightarrow{\partial_n} \cdots \xrightarrow{\partial_1} C_0(\tilde{X}) \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0$$

is exact. The boundary maps are $\mathbb{Z}G$ -module homomorphisms. This gives a free resolution of $\mathbb{Z}$ by free $\mathbb{Z}G$ -modules.

Theorem 2.10. *There is a unique resolution of $\mathbb{Z}$ by free (projective) $\mathbb{Z}G$ -module.*

Definition 2.11 (Cohomology of group). Let M be a left $\mathbb{Z}G$ -module. Let $F = \{(F_i, d_i)\}$ be a free resolution of $\mathbb{Z}$. We truncate F to get a (non-exact) complex:

$$\cdots \rightarrow F_i \xrightarrow{d_i} F_{i-1} \xrightarrow{d_{i-1}} \cdots \xrightarrow{d_1} F_0 \rightarrow 0.$$

Apply $\text{Hom}_{\mathbb{Z}G}(-, M)$ to the complex. Denote $F^i = F_i^* = \text{Hom}_{\mathbb{Z}G}(F_i, M)$. We have:

$$\cdots \leftarrow F^i \xleftarrow{d^{i-1}} F^{i-1} \xleftarrow{d^{i-2}} \cdots \xleftarrow{d^0} F^0 \leftarrow 0.$$

The *cohomology of G* with coefficients in M , denoted as $H^*(G; M)$, is the homology of the complex above:

$$H^i(G; M) = \ker d^i / \text{im } d^{i-1}.$$

If we do not truncate the resolution, then we get the exact sequence:

$$\cdots \leftarrow \text{Hom}_{\mathbb{Z}G}(F_1, M) \xleftarrow{d^0=d_1^*} \text{Hom}_{\mathbb{Z}G}(F_0, M) \xleftarrow{\epsilon^*} \text{Hom}_{\mathbb{Z}G}(\mathbb{Z}, M) \leftarrow 0.$$

Denote $H^0(G; M)$ as M^G . Notice that $M^G = \ker d^0 = \text{im } \epsilon^* \cong \text{Hom}_{\mathbb{Z}G}(\mathbb{Z}, M)$. Recall that G acts on $\mathbb{Z}$ trivially, so elements in $\text{Hom}_{\mathbb{Z}G}(\mathbb{Z}, M)$ have to be elements in M that are invariant under G , i.e. $M^G = \{m \in M \mid \forall g \in G, gm = m\}$.

Since free resolutions are chain homotopic, $H^*(G; M)$ is independent of the choices of the free resolution.

Example 2.12 (Cohomology of $\mathbb{Z}/n\mathbb{Z}$). Let $G = \mathbb{Z}/n\mathbb{Z}$, $M = \mathbb{Z}$. Then $\mathbb{Z}G = \mathbb{Z}[t]/(t^n - 1)$. Consider $\mathbb{Z}G \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0$, $\ker \epsilon = (t - 1)$. Let $N = t^{n-1} + \cdots + t + 1$. We have the free resolution:

$$\cdots \rightarrow \mathbb{Z}G \xrightarrow{\cdot N} \mathbb{Z}G \xrightarrow{\cdot (t-1)} \mathbb{Z}G \xrightarrow{\cdot N} \mathbb{Z}G \xrightarrow{\cdot (t-1)} \mathbb{Z}G \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0.$$

Apply $\text{Hom}_{\mathbb{Z}G}(-, \mathbb{Z})$ to the resolution. For any $f \in \text{Hom}_{\mathbb{Z}G}(\mathbb{Z}G, \mathbb{Z})$, $f(\sum_i \alpha_i t^i)$ has to be $k \sum_i \alpha_i$ where $k \in \mathbb{Z}$, i.e. $\text{Hom}_{\mathbb{Z}G}(\mathbb{Z}G, \mathbb{Z}) = \mathbb{Z}$. Thus the cochain complex we obtained is:

$$\cdots \xleftarrow{n} \mathbb{Z} \xleftarrow{0} \mathbb{Z} \xleftarrow{n} \mathbb{Z} \xleftarrow{0} \mathbb{Z} \leftarrow 0.$$

We conclude that

$$H^i(\mathbb{Z}/n\mathbb{Z}; \mathbb{Z}) = \begin{cases} \mathbb{Z} & \text{if } i = 0, \\ \mathbb{Z}/n\mathbb{Z} & \text{if } i \geq 1 \text{ is odd,} \\ 0 & \text{if } i \geq 1 \text{ is even.} \end{cases}$$

Proposition 2.13. *If X is an n -dimensional $K(G, 1)$, M is any $\mathbb{Z}G$ -module, then $\forall m > n$, $H^m(G; M) = 0$.*

Proof. The free resolution is trivial for $m > n$. □

Corollary 2.14. *There does not exist any finite-dimensional $K(\mathbb{Z}/n\mathbb{Z}, 1)$ for $n \geq 2$.*

We are now ready to prove Theorem 1.10. Recall the statement:

Theorem 1.10. *If G has non-trivial torsion, then there does not exist any finite-dimensional $K(G, 1)$.*

Proof. Let X be a $K(G, 1)$. For any subgroup $H \leq G$, let X^H be the cover of X corresponding to H . Then X^H is a $K(H, 1)$. Observe that X and X^H have the same dimension.

If G has non-trivial torsion, then there exists $\mathbb{Z}/n\mathbb{Z} \leq G$ with $n \geq 2$. By Corollary 2.14, X^H is infinite-dimensional, hence so is X . $\square$

3. COMPACTLY SUPPORTED CELLULAR COHOMOLOGY

In this course we focus on $\mathbb{Z}G$ -module with trivial coefficient. For coefficients other than $\mathbb{Z}$, a key phrase is “local systems of coefficients on X ”.

Theorem 3.1. *If X is a $K(G, 1)$, then for all $i \geq 0$, $H^i(G; \mathbb{Z}) \cong H^i(X; \mathbb{Z})$.*

Proof. Recall that X gives a free $\mathbb{Z}G$ resolution of $\mathbb{Z}$ (See Remark 2.9):

$$\cdots \rightarrow C_1(\tilde{X}) \xrightarrow{d_1} C_0(\tilde{X}) \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0.$$

Apply $\text{Hom}_{\mathbb{Z}G}(\cdot, \mathbb{Z})$ to the resolution and consider $\text{Hom}_{\mathbb{Z}G}(C_i(\tilde{X}), \mathbb{Z})$. Recall that $\mathbb{Z}G$ homomorphism has to be invariant under G , so the map has to assign the same number to each i -cell in the same G -orbit. It is naturally equivalent to assigning numbers to i -cells in X i.e. cellular chains in X . $\square$

Definition 3.2 (Compactly supported homomorphism). Let M be a $\mathbb{Z}G$ -module. Define $\text{Hom}_c(M, \mathbb{Z}) \subseteq \text{Hom}(M, \mathbb{Z})$ to be

$$\{f : M \rightarrow \mathbb{Z} \mid \forall m \in M, \text{ for all but finitely many } g \in G, f(gm) = 0\}.$$

Lemma 3.3 ([Bro82, Lemma VIII.7.4]). *Let Γ be a $\mathbb{Z}G$ -module, then*

$$\text{Hom}_{\mathbb{Z}G}(M, \mathbb{Z}G) \cong \text{Hom}_c(M, \mathbb{Z}).$$

Proof. Take $F \in \text{Hom}_{\mathbb{Z}G}(M, \mathbb{Z}G)$. For $m \in M$, $F(m)$ can be written as $\sum_{g \in G} f_g(m)g$ where $f_g(m) = 0$ for all but finitely many g .

Verify that $F \in \text{Hom}_{\mathbb{Z}G}(M, \mathbb{Z}G)$ if and only if $f_g(m) = f_1(g^{-1}m)$. So $F \mapsto f_1$ gives a map whose inverse is:

$$f \mapsto \{m \mapsto \sum_{g \in G} f(g^{-1}m)g\}.$$

$\square$

Definition 3.4 (Compactly supported cohomology). Let Y be a locally compact CW complex, i.e. for all n -cell σ , there are finitely many $n+1$ cells whose boundary meets σ . Let $C_c^i(Y; \mathbb{Z}) \subseteq C^i(Y; \mathbb{Z})$ be the set of finitely supported cellular cochains with

$$\delta_c^i : C_c^i(Y; \mathbb{Z}) \rightarrow C_c^{i+1}(Y; \mathbb{Z}).$$

The homology of this chain complex, denoted as $H_c^i(Y; \mathbb{Z})$, is the *compactly supported cohomology*.

Remark 3.5. The requirement of locally compactness is to ensure that finitely supported cochains and compactly supported cochains are the same set.

Theorem 3.6. *If X is a finite $K(G, 1)$, then*

$$H^i(G; \mathbb{Z}G) \cong H_c^i(\tilde{X}; \mathbb{Z}).$$

Proof. Recall that by Lemma 3.3,

$$\text{Hom}_{\mathbb{Z}G}(C_i(\tilde{X}), \mathbb{Z}G) \cong \text{Hom}_c(C_i(\tilde{X}), \mathbb{Z}) = C_c^i(\tilde{X}; \mathbb{Z}).$$

So they induce the same cohomology (See Definitions 2.11 and 3.4). $\square$

Remark 3.7. A more natural definition of compactly supported cohomology arises from the direct limit of relative cohomology groups. Observe that a finitely supported i -cochain is 0 outside some compact set K , so it lies in $C^i(Y, Y \setminus K; \mathbb{Z})$. We define:

$$H_c^*(Y; \mathbb{Z}) = \varinjlim_{K \text{ compact}} H^*(Y, Y \setminus K; \mathbb{Z}).$$

We give some examples of calculating group cohomology using compactly supported cohomology below.

Observe that for a connected infinite CW complex Y , $H_c^0(Y; \mathbb{Z}) = \ker \delta_c^0$, which is the set of finitely supported 1-cochains that are constant. So it has to be trivial. (It would be $\mathbb{Z}$ if Y is finite.)

Example 3.8 (Compactly supported cohomology of $\mathbb{R}$). Let $\mathbb{R}$ be cellulated as an infinite graph. Let v_i be the vertices and let $e_i = \{v_i, v_{i+1}\}$ be the edges.

Since $\mathbb{R}$ is contractible, $H_c^0(\mathbb{R}; \mathbb{Z}) = 0$. For $n > 2$, since there are no n -cells, $H_c^n(\mathbb{R}; \mathbb{Z}) = 0$. Consider now $H_c^1(\mathbb{R}; \mathbb{Z})$. Since there are no 2-cells, all 1-cochains are 1-cocycles. Therefore $H_c^1(\mathbb{R}; \mathbb{Z}) = C_c^1(\mathbb{R}; \mathbb{Z}) / \text{im } \delta_c^0$.

Let v_i^*, e_i^* be the dual cells of e_i, v_i respectively. Consider the augmented chain $C_c^0(\mathbb{R}; \mathbb{Z}) \xrightarrow{\delta_c^0} C_c^1(\mathbb{R}; \mathbb{Z}) \xrightarrow{\epsilon} \mathbb{Z}$ where ϵ takes every e_i^* to 1. Take $\alpha \in C_c^0(\mathbb{R}; \mathbb{Z})$. We can write α as $\sum_i a_i v_i^*$. Then $\delta_c^0(\alpha) = \sum_i a_i (\delta_c^0(v_i^*))$. But $\delta_c^0(v_i^*) = e_i^* - e_{i+1}^*$, so $\epsilon(\delta_c^0(\alpha)) = \sum_i a_i (1 - 1) = 0$. That is, $\text{im } \delta_c^0 \subseteq \ker \epsilon$. As a result, ϵ factors through $\bar{\epsilon} : C_c^1(\mathbb{R}; \mathbb{Z}) / \text{im } \delta_c^0 = H_c^1(\mathbb{R}; \mathbb{Z}) \rightarrow \mathbb{Z}$.

Clearly $\bar{\epsilon}$ is surjective. To show that $\bar{\epsilon}$ is injective, we need to prove that if $\sum_{i=m}^n a_i = 0$, then $\sum_{i=m}^n a_i e_i^* \in \text{im } \delta_c^0$. We remark that it's important that $\sum_{i=m}^n a_i$ is a finite sum. Notice that

$$\cdots + a_{n-1} e_{n-1}^* + a_n e_n^* + a_n \delta_c^0(v_{n-1}^*) = \cdots + (a_{n-1} + a_n) e_{n-1}^* + 0 \cdot e_n^*,$$

so by adding some coboundary consecutively, we can get a cochain supported on a single edge with coefficient zero. Thus the original cochain is also a coboundary.

Corollary 3.9.

$$H_c^i(\mathbb{Z}; \mathbb{Z}[\mathbb{Z}]) = \begin{cases} \mathbb{Z} & i = 1, \\ 0 & \text{otherwise.} \end{cases}$$

Example 3.10 (Compactly supported cohomology of infinite 4-valence tree). Consider the infinite 4-valence tree T (the universal cover of the wedge of two circles). Similarly, $H_c^i(T; \mathbb{Z}) = 0$ when $i \neq 1$. For $H_c^1(T; \mathbb{Z})$, by Remark 3.7 it is the direct limit of $H^1(T, T \setminus K; \mathbb{Z})$ over all compact sets. Consider the long exact sequence of reduced cohomology groups:

$$\tilde{H}^0(T; \mathbb{Z}) \rightarrow \tilde{H}^0(T \setminus K; \mathbb{Z}) \rightarrow \tilde{H}^1(T, T \setminus K; \mathbb{Z}) \rightarrow \tilde{H}^1(T; \mathbb{Z}).$$

Since T is contractible, $\tilde{H}^i(T; \mathbb{Z}) = 0$, hence $\tilde{H}^0(T \setminus K; \mathbb{Z}) \cong \tilde{H}^1(T, T \setminus K; \mathbb{Z})$. Now $\tilde{H}^0(T \setminus K; \mathbb{Z})$ counts the components of $T \setminus K$, which increases infinitely as the size of K increases.

We conclude that $\tilde{H}^1(T, T \setminus K; \mathbb{Z})$ is infinitely generated. In fact, it's an infinitely generated free abelian group.

Corollary 3.11.

$$H_c^i(\mathbb{F}_2; \mathbb{Z}[\mathbb{F}_2]) = \begin{cases} D & i = 1, \\ 0 & \text{otherwise,} \end{cases}$$

where D is infinitely generated.

Observe that the argument of Example 3.10 works for any contractible space. That is, in general, 1-cohomology of a group with group-ring coefficients has rank of numbers of ends of the group plus 1.

4. GROUP COHOMOLOGY OF SURFACE GROUPS

Example 4.1 (Compactly supported cohomology of R^n). Suppose $\mathbb{R}^n$ is equipped with a polyhedral decomposition. Suppose each $(n-1)$ -cell is adjacent to 2 oriented n -cells, and the $n-1$ -cell appears in their boundary with opposite orientation. Suppose further that each compact $K \subseteq \mathbb{R}^n$ is contained in a ball. (For example, regard R^3 as the universal covering space of 3-torus.)

Since $\mathbb{R}^n$ is contractible, $H_c^0(\mathbb{R}^n; \mathbb{Z}) = 0$.

For $1 \leq i < n$, take $\eta \in C_c^i(\mathbb{R}^n; \mathbb{Z})$ such that $\delta\eta = 0$. Take a ball B such that $\text{supp}(\eta) \subseteq B$. We can regard η as an element of $H^i(\mathbb{R}^n, \mathbb{R}^n \setminus B)$ (that is, a cochain on $\mathbb{R}^n$ that vanishes on B). Recall that $H^i(\mathbb{R}^n, \mathbb{R}^n \setminus B) \cong \tilde{H}^{i-1}(\mathbb{R}^n \setminus B) \cong \tilde{H}^{i-1}(\mathbb{S}^{n-1}) = 0$.

Consider now the augmented chain $C_c^{n-1}(\mathbb{R}^n; \mathbb{Z}) \xrightarrow{\delta} C_c^n(\mathbb{R}^n; \mathbb{Z}) \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0$. The assumption that each $(n-1)$ -cell occurs in the boundary of two n -cells with opposite orientation implies that $\text{im } \delta \subseteq \ker \epsilon$ (similar to Example 3.8), so ϵ factors through $H_c^n(\mathbb{R}^n; \mathbb{Z}) \rightarrow \mathbb{Z}$. Again, we need to show that the map is injective. The argument is similar to Example 3.8: we take the dual graph of $\mathbb{R}^n$ so that vertices are dual n -cells and edges are dual $(n-1)$ -cells. We then adjust one dual n -cell at a time until the cochain is supported on a single dual n -cell with coefficient zero.

Corollary 4.2. *Let M be a closed n -manifold with $\widetilde{M} \cong \mathbb{R}^n$. Let $G = \pi_1(M)$. Then*

$$H^i(G; \mathbb{Z}G) \cong H_c^i(\mathbb{R}^n; \mathbb{Z}) = \begin{cases} \mathbb{Z} & i = n, \\ 0 & \text{otherwise.} \end{cases}$$

In particular, if Σ_g is a closed orientable surface with genus $g \geq 1$ and $G = \pi_1(\Sigma_g)$, then

$$H^i(G; \mathbb{Z}G) = \begin{cases} \mathbb{Z} & i = 2, \\ 0 & \text{otherwise.} \end{cases}$$

Remark 4.3. For non-orientable manifolds, the cohomology would be the same. What's different is the action of Γ on the top cohomology $\mathbb{Z}$.

In general, $H^i(G; \mathbb{Z}G)$ is a $\mathbb{Z}G$ -module. Take a free resolution F . A priori, $\text{Hom}_{\mathbb{Z}G}(F_i, \mathbb{Z}G)$ is just an abelian group, but the right action by G satisfies the criterion from Lemma 3.3. Therefore, as a quotient of $\mathbb{Z}G$ -module by $\mathbb{Z}G$ -module, $H^i(G; \mathbb{Z}G)$ is also a $\mathbb{Z}G$ -module.

Question 4.4 (Wall). Suppose G has type F (recall Definition 1.11) and suppose that for some n , $H^n(G; \mathbb{Z}G) = \mathbb{Z}$ and $H^i(G; \mathbb{Z}G) = 0$ otherwise. Does there exist a closed aspherical manifold M such that $G = \pi_1(M)$?

Question 4.4 is known to be true for $n = 1$ ($G = \mathbb{Z}$) and $n = 2$ (surface groups). It is wide open for $n \geq 3$.

Notice that if such an M exists, it's unique up to homotopic equivalence by the uniqueness of $K(G, 1)$. It's natural to ask:

Question 4.5 (Borel conjecture). If M_1, M_2 are closed aspherical manifolds that are homotopic equivalent to each other, are they homeomorphic to each other?

Question 4.5 is known to be true for $n = 1, 2, 3$. Note that in general, without the assumption of aspherical, this is not true.

Recall that two homotopic equivalent manifolds M_1, M_2 have isomorphic fundamental groups. A group theoretic version of Question 4.5 is: Given a group G , does it have a unique $K(G, 1)$ up to homeomorphism?

Theorem 4.6. *Let X be a CW complex such that $\tilde{X}$ is k -connected (recall Definition 2.1) for some k . Then there exists an aspherical CW complex Z such that $Z^{k+1} = X^{k+1}$.*

Theorem 4.6 is the reason we are interested in 1-connected spaces, because we can obtain a CW complex with the same fundamental group.

Proof. Assume inductively that we've built Z_n such that $\tilde{Z}_n$ is n -connected and $Z_n^{k+1} = X^{k+1}$ (Z_k is just X). Note that every map $f : \mathbb{S}^{n+1} \rightarrow Z_n$ is homotopic to a map with image in Z_n^{n+1} . Let $\{F_\alpha : \mathbb{S}^{n+1} \rightarrow Z_n^{n+1} \mid \alpha \in A_n\}$ denote the homotopy classes of maps $\mathbb{S}^{n+1} \rightarrow Z_n$. Build Z_{n+1} by gluing $(n+1)$ -cells to Z_n using F_α s as attaching maps.

Observe that $Z_{n+1}^{n+1} = Z_n^{n+1}$. Any $g : \mathbb{S}^{n+1} \rightarrow Z_{n+1}$ is homotopic to some F_α , hence is null-homotopic. Therefore $\tilde{Z}_{n+1}$ is $(n+1)$ -connected. $\square$

5. PROJECTIVE MODULES AND MORE FINITENESS PROPERTIES

Recall Theorem 3.6. In fact, to obtain the isomorphism $H^i(G; \mathbb{Z}G) \cong H_c^i(\tilde{X}; \mathbb{Z})$, we need only finite X^{i+1} .

Definition 5.1. Let G be a group. The *cohomological dimension* of G , denoted as $\text{cd}(G)$ (or $\text{cd}_{\mathbb{Z}}(G)$), is defined to be

$$\sup\{n \mid \exists M \text{ a } \mathbb{Z}G\text{-module such that } H^n(G; M) \neq 0\}.$$

If there's no bound on n , then we say $\text{cd}(G) = \infty$.

Recall geometric dimension of a group (See Definition 1.11). Given a $K(G, 1)$ of dimension n , we obtain a free resolution of $\mathbb{Z}$ by $\mathbb{Z}G$ -modules that stops in degree n . Consider its cohomology: there is no cochain groups in dimension above n , so the cohomology groups in dimension above n are trivial. That is, the geometric dimension is an upper bound of the cohomological dimension.

Question 5.2. Are there any G with $\text{cd}(G) < \text{gd}(G)$?

Theorem 5.3 ([EG57]). *Let G be a group. Let $k = \max\{\text{cd}(G), 3\}$. Then there exists a $K(G, 1)$ of dimension k .*

So if $\text{cd}(G) \geq 3$, then $\text{cd}(G) = \text{gd}(G)$.

In lower dimensions, $\text{cd}(G) = 0$ if and only if G is trivial if and only if $\text{gd}(G) = 0$. (This is a good exercise). At dimension 1, we have $\text{cd}(G) = 1$ if and only if G is free if and only if $\text{gd}(G) = 1$. However, the proof is significantly harder.

Theorem 5.4 (Stallings-Swan theorem). *If $\text{cd}(G) = 1$ then G is free.*

The case at 2 is still an open question, though there is already a potential counterexample. TODO add reference to the potential counterexample.

Definition 5.5. A $\mathbb{Z}G$ -module P is *projective* if for every exact

$$M_0 \xrightarrow{i} M_1 \xrightarrow{j} M_2,$$

and every map $\phi : P \rightarrow M_1$ with $j\phi = 0$, there exists $\psi : P \rightarrow M_0$ such that $i\psi = \phi$.

$$\begin{array}{ccccc} & & P & & \\ & \swarrow \psi & \downarrow \phi & \searrow 0 & \\ M_0 & \xrightarrow{i} & M_1 & \xrightarrow{j} & M_2 \end{array}$$

Lemma 5.6. *Free modules are projective.*

Proof. Trace the basis of the free module. □

Theorem 5.7. *Projective resolutions are unique up to chain homology.*

Lemma 5.8. *Let P be a projective module and let Q be a direct summand of P . Then Q is projective.*

Proof. Suppose $P = Q \oplus M$.

$$\begin{array}{ccccc} & & P = Q \oplus M & & \\ & \swarrow \psi_P & \uparrow \iota \downarrow \pi & \searrow 0 & \\ & & Q & & \\ & \swarrow \psi_P \iota & \downarrow \phi & \searrow 0 & \\ M_0 & \xrightarrow{i} & M_1 & \xrightarrow{j} & M_2 \end{array}$$

□

Corollary 5.9. *Direct summands of free modules are projective.*

Proposition 5.10. *TFAE:*

- (1) P is projective.
- (2) For every surjective $\pi : M \rightarrow \overline{M}$ and every $\phi : P \rightarrow \overline{M}$, there exists $\psi : P \rightarrow M$ such that $\pi\psi = \phi$.

- (3) *Every exact $0 \rightarrow M \rightarrow M' \rightarrow P \rightarrow 0$ splits.*
 (4) *P is a direct summand of a free module.*

Proof. (1) $\implies$ (2): Replace M by $\ker j$ in Definition 5.5.

(2) $\implies$ (3): Lift $\text{Id} : P \rightarrow P$ to $P \rightarrow M'$.

(3) $\implies$ (4): Exhibits P as a quotient of a free module, then the following SES splits:

$$0 \rightarrow \ker \epsilon \rightarrow F \xrightarrow{\epsilon} P \rightarrow 0.$$

(4) $\implies$ (1): Corollary 5.9. □

Definition 5.11. Let G be a group.

We say G has type FF (resp. FP) if there exists a finite resolution of $\mathbb{Z}$ by finitely generated free (resp. projective) $\mathbb{Z}G$ -modules.

We say G has type FF_n (resp. FP_n) if there exists a resolution

$$\cdots \rightarrow M_1 \rightarrow M_0 \rightarrow \mathbb{Z} \rightarrow 0$$

of $\mathbb{Z}$ by free (resp. projective) $\mathbb{Z}G$ -modules with finitely-generated M_i for $i \leq n$.

We say G has type FF_∞ (resp. FP_∞) if G has type FF_n (resp. FP_n) for all n .

Theorem 5.12. *Type FF implies type FP .*

Question 5.13 (Serre's Conjecture). Does type FP imply type FF ?

Recall type F, F_n, F_∞ (see Definition 1.11). Clearly $F_n \implies FF_n \implies FP_n$. Recall also Theorem 1.9.

Theorem 5.14. *There exists G of type FP_2 which is not finitely presented (hence is not of type F_2)*

Right-angled Artin groups is a class of groups that can tells those properties apart.

Definition 5.15 (Right-angled Artin group). Let Γ be a simplicial graph with vertex set V and edge set E . We associate a group A_Γ to Γ

$$A_\Gamma := \langle V \mid vw = wv \text{ if } \{v, w\} \in E \rangle.$$

Example 5.16. (1) If $|V| = m$ and $E = \emptyset$, then $A_\Gamma = \mathbb{F}_m$.

(2) If $|V| = m$ and Γ is a complete graph, then $A_\Gamma = \mathbb{Z}^m$.

(3) Say $V = \{a, b, c\}$ and $E = \{\{a, b\}\}$, then $A_\Gamma = \mathbb{Z}^2 * \mathbb{Z}$.

Theorem 5.17. *Let k be the clique number (i.e. maximal size of complete subgraph) of Γ . Then $\text{gd}(A_\Gamma) = k$.*

Remark 5.18. Notice that under the assumption of Theorem 5.17, $\mathbb{Z}^k \subseteq A_\Gamma$, so $\text{cd}(A_\Gamma) \geq k$. We are interested in the subgroups of A_Γ and we will investigate the map $\phi_\Gamma : A_\Gamma \rightarrow \mathbb{Z}$, defined by $v \mapsto 1, \forall v \in V$.

6. THE BESTVINA-BRADY THEOREM FOR FLAG COMPLEXES

Definition 6.1 (flag simplicial complex). A simplicial complex Γ is *flag* if whenever a set of vertices $\{v_0, \dots, v_k\}$ span a complete subgraph, then they span a k -simplex.

In other words, every non-simplex contains a non-edge.

Any graph Γ determines a flag simplicial complex.

Definition 6.2 (RAAGs of flag simplicial complex). Let Γ be a simplicial graph with vertex set V and edge set E . Let A_Γ be the associated RAAG (See Definition 5.15). If L is a flag simplicial complex, Then $A_L := A_{L(1)}$.

Define $\phi_L : A_L \rightarrow \mathbb{Z}$ by $v \mapsto 1$. Let $K_L = \ker \phi_L$.

Remark 6.3. Since ϕ_L is a homomorphism, K_L is related to the “topology” of flag of Γ .

Example 6.4. Let L be a square, then $A_L = \mathbb{F}_2 \times \mathbb{F}_2$.

In general, join of graphs corresponds to product of groups. (Disjoint union corresponds to free product).

Definition 6.5. A space X is *acyclic* if $\tilde{H}_i(X) = 0$ for all $i \geq 0$.

Recall that a group is *perfect* if it has trivial abelianization, i.e. $\tilde{H}_1(X)$ is trivial. By definition, every acyclic group is perfect.

Definition 6.6. A space X is *homologically (k) -connected* if $\tilde{H}_i(X) = 0$ for $i \leq k$

Theorem 6.7 (The Bestvina-Brady Theorem). *Let L be a finite flag simplicial complex.*

- (1) K_L is finitely presented if and only if L is simply connected.
- (2) K_L has type FP if and only if L is acyclic.
- (3) K_L has type FP_n if and only if L is homologically $(n-1)$ -connected.

Theorem 6.8 ([Hat01, Theorem 4.32]). *If X is $(n-1)$ -connected, $n \geq 2$, then $\tilde{H}_i(X) = 0$ for $0 \leq i \leq n-1$, and $H_n(X) \cong \pi_n(X)$.*

Theorem 6.8 will guide our search for a non-contractible acyclic space. We also want π_1 to be perfect.

Example 6.9 (Poincare’s homology sphere). Take an empty dodecahedron such that opposite sides are the same up to a twist. Identify opposite sides with a $\pi/5$ twist, then we get the 2-skeleton of Poincare’s homology sphere, denoted as X .

Let M be the homology sphere (which is a 3-manifold, and has X as its 2-skeleton). We have $\tilde{H}_3(M) \cong \tilde{H}_3(\mathbb{S}^3) = \mathbb{Z}$, and that $\tilde{H}_i(M) \cong \tilde{H}_i(\mathbb{S}^3) = 0$ for all other i ’s.

Recall Universal coefficient: if $H_i(M)$ is torsion free, then $H^{i+1}(M; \mathbb{Z}) \cong \text{Hom}(H_i(M), \mathbb{Z})$.

Recall Poincare Duality: $H_i(M) \cong H^{3-i}(M; \mathbb{Z})$

Since $H_0(M) = 0$, $H_1(M) = 0 \implies H^1(M; \mathbb{Z}) = 0$ by Universal Coefficient theorem, then $H_2(M) = 0$ (by Poincare Duality).

So do Cellular chain complex for M we get

$$0 \rightarrow C_3(M) \xrightarrow{\alpha_3} C_2(M) \rightarrow C_1(M) \rightarrow \mathbb{Z} \rightarrow 0$$

Observe that $\alpha_3 = 0$ because M has no boundary after gluing. Also $C_3(M) = \mathbb{Z}$ because there's only one 3-cell.

We can ignore $C_3(M)$. The rest of the chain (and the homology) coincides with that of X :

$$0 \rightarrow C_2(X) \rightarrow C_1(X) \rightarrow \mathbb{Z} \rightarrow 0$$

And so all homologies of X are trivial.

This X with K_X gives the first example of a group which is acyclic but non-contractible complex.

Now subdivide the faces to (10) triangles to obtain a flag complex L . This L is not simply connected, so K_L is not finitely presented. It is still acyclic, so K_L has type FP . This gives the first example of a group that has FP but is not finitely presented.

Remark 6.10. If we do $2\pi/5$ rotation with filled dodecahedron, then we get an infinite perfect π_1 , and it's hyperbolic.

One can see that by seeing how there are 5 pentagons around an edge, so angle argument gives the hyperbolic geometry.

Our goal now is to prove Theorem 6.7. We first introduce yet another finiteness property.

Definition 6.11. A group G has type FH_n (resp. FH) if there exists a homologically $(n-1)$ -connected (resp. acyclic) complex X with a free cellular cocompact G action.

We now have four types of finiteness properties: F, FP, FF, FH (See Definitions 1.11, 5.11 and 6.11).

We know that $F_n \implies FH_n \implies FF_n \implies FP_n$. We know there exists a group that has type FP_2 but not type F_2 (Recall Theorem 5.14). Lemma 6.12 proves $FP_n \implies FF_n$.

Similarly, $F \implies FH \implies FF \implies FP$ (The last part is Theorem 5.12). It's still an open question whether $FP \implies FF$ (See Question 5.13).

Lemma 6.12. $FP_n \implies FF_n$.

Proof. Let M be a projective $\mathbb{Z}G$ -module. Take M as a quotient of free module:

$$0 \rightarrow K \rightarrow F \rightarrow M \rightarrow 0.$$

If M is finitely generated, we can assume F is also finitely generated. By Proposition 5.10 K is projective and the sequence splits, which forces K to be finitely generated. Hence $F = K \oplus M$.

Take a finitely generated projective resolution $\{P_i\}_{i=0}^n$ of $\mathbb{Z}$. Then there exists $F_0 = P_0 \oplus Q_0$. Build:

$$P_1 \oplus Q_0 \rightarrow P_0 \oplus Q_0 \rightarrow \mathbb{Z}$$

As a direct sum of projective modules, $P_1 \oplus Q_0$ is also projective. We can then find Q_1 such that $Q_1 \oplus P_1 \oplus Q_0$ is a finitely generated free module. We can repeat this procedure to eventually obtain a finitely generated free resolution. $\square$

We make some observations below.

- (1) If there exists $n-1$ -connected X with a free cocompact G action, then G has type F_n by turning $G \backslash X$ into a $K(G, 1)$. This proves $F_n \implies FH_n$.
- (2) $FH_n \implies FF_n$ Take the X you get, $C_i(X)$ are finitely generated-free $\mathbb{Z}G$ -modules, and this is exact because it's homologically n -connected.
- (3) $FH_1 \implies F_1$ because 0th homology is about being connected.
- (4) $FF_1 \implies F_1$ See Lemma 7.4.

In particular, we have:

$$F_1 \iff FF_1 \iff FH_1 \iff \text{finitely generated.}$$

7. SCHANUEL'S LEMMA AND THE SALVETTI COMPLEX

Lemma 7.1 (Schanuel's Lemma). *Suppose P, P' are projective modules, and we have exact sequences*

$$0 \rightarrow K \rightarrow P \xrightarrow{\pi} M \rightarrow 0,$$

$$0 \rightarrow K' \rightarrow P' \xrightarrow{\pi'} M \rightarrow 0.$$

Then $P \oplus K' \cong P' \oplus K$.

Proof. Let $Q = \{(p, p') \in P \oplus P' \mid \pi(p) = \pi'(p')\}$ be the fiber product of π, π' . Define $\alpha : Q \rightarrow P$ by $(p, p') \mapsto p$. Then $\ker \alpha = \{(p, p') \mid p = 0, \pi'(p') = \pi(p) = 0\}$. But that's just $0 \oplus \ker \pi' = K'$. Since π' is surjective, so $\forall p \in P, \exists p' \in P'$ such that $\pi'(p') = \pi(p)$, that is, α is surjective. So the following short sequence is exact:

$$0 \rightarrow K' \rightarrow Q \xrightarrow{\alpha} P \rightarrow 0.$$

Since P is projective, $Q \cong P \oplus K'$ (See Proposition 5.10). By the same argument, $P' \oplus K \cong Q$. $\square$

Corollary 7.2. *Suppose $P_1 \rightarrow P_0 \xrightarrow{\pi} M \rightarrow 0$ is an exact complex where P_0, P_1 are finitely generated projective modules. Suppose also $P'_0 \xrightarrow{\pi'} M \rightarrow 0$ is an exact complex where P'_0 is a finitely generated projective module. Then $\ker \pi'$ is finitely generated.*

Proof. The map π gives a short exact sequence:

$$0 \rightarrow \ker \pi \rightarrow P_0 \xrightarrow{\pi} M \rightarrow 0.$$

Notice that as a submodule of a finitely generated module, $\ker \pi$ is also finitely generated.

We also have:

$$0 \rightarrow \ker \pi' \rightarrow P'_0 \xrightarrow{\pi'} M \rightarrow 0.$$

By Lemma 7.1, $\ker \pi \oplus P'_0 \cong \ker \pi' \oplus P_0$. Since $\ker \pi$, P'_0 , P_0 are all finitely generated, so is $\ker \pi'$. $\square$

Recall that groups with type FP_n give rise to finitely generated projective resolutions (See Definition 5.11). Proposition 7.3 is a higher dimension version of Corollary 7.2.

Proposition 7.3. *Let M be a module. Suppose $\{P_i\}_{i=0}^n$ is a finitely generated projective resolution of M . Suppose $\{P'_i\}_{i=0}^k$ is another finitely generated projective resolution and $k < n$. Then $\ker(P'_k \rightarrow P'_{k-1})$ is also finitely generated.*

Recall Definition 5.11 and Definition 1.11.

Lemma 7.4. *Suppose a group G has type FF_1 , then G is finitely generated.*

Proof. If a group G has type FF_1 then there exists a free resolution of $\mathbb{Z}$ with $\mathbb{Z}G$ -module:

$$F_1 \rightarrow F_0 \xrightarrow{\alpha} \mathbb{Z} \rightarrow 0.$$

We also have the map

$$\mathbb{Z}G \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0, \quad g \mapsto 1.$$

By Corollary 7.2, $\ker \epsilon$ is finitely generated as a $\mathbb{Z}G$ -module. FIX?

Claim 1. Group G is finitely generated if and only if $\ker \epsilon$ is finitely generated as a $\mathbb{Z}G$ -module.

Let S be a generating set of G , then $\{1 - s \mid s \in S\}$ is a finite generating set of $\ker \epsilon$. To see this, notice two facts:

- (1) $\ker \epsilon$ is generated (as an abelian group) by $\{1 - g \mid g \in G\}$, and
- (2) $1 - ab = a(1 - b) + (1 - a)$, that is any $1 - g$ can be written as combinations of $1 - s$ for some $s \in S$.

Suppose now $\ker \epsilon$ is finitely generated. Say it's generated by $\{1 - s \mid s \in S\}$ for some finite set S . Take the Cayley graph Γ of G with respect to S , recall that Γ is connected if and only if S generates G . Now consider the chain complex

$$C_1(\Gamma) \xrightarrow{\partial_1} C_0(\Gamma) = \mathbb{Z}G \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0.$$

We verify that $\partial_1(e) = g - gs = g(1 - s) = 0$, so $\text{im } \partial_1$ is a $\mathbb{Z}G$ module generated by $\{1 - s\}$, which is exactly $\ker \epsilon = \text{im } \partial_1$.

It follows that $\tilde{H}_0(\Gamma) = 0$, i.e. Γ is connected, i.e. G is finitely generated. $\square$

Proposition 7.5. *If G has type FF_2 , then G has type FH_2 . (See Definitions 5.11 and 6.11.)*

Proof. Suppose G has type FF_2 , then G has type FF_1 and is finitely generated by Lemma 7.4. Suppose $G = \langle S | R \rangle$ where S is finite. If R is finite, then we are done. If not, we build a presentation 2-complex Z and let $\tilde{Z}$ be the universal cover. We then have:

$$C_2(\tilde{Z}) \xrightarrow{\partial_2} C_1(\tilde{Z}) \xrightarrow{\partial_1} C_0(\tilde{Z}) \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0,$$

where $C_1(\tilde{Z}), C_0(\tilde{Z})$ are finitely generated free $\mathbb{Z}G$ -module (generated by orbits of i -cells in Z). By an upgraded version of Lemma 7.1, $\ker \partial_1$ is a finitely generated $\mathbb{Z}G$ -module. So there exists a finite subset of 2-cells in Z such that if we add only their orbits to $\tilde{Z}^{(1)}$, we get a space X_0 with finitely many orbits of 2-cells. So X_0 is homologically 1-connected. (Note that X_0 is probably not simply-connected). $\square$

Question 7.6 (Bestvina-Brady). Does FF_n imply FH_n for $n > 2$?

Definition 7.7. Let L be a finite flag simplicial complex and let A_L be the RAAG associated to L . The Salvetti complex is a presentation 2-complex X_L built in the following way:

- X_L has one vertex x_0 ,
- For each $v \in V(L)$, add a loop in X_L
- For each $\{v, w\} \in E(L)$, associate a torus in X_L , whose boundaries correspond to v, w .
- For each simplex in L spanned by $\{v_0, \dots, v_k\}$, add a $(k + 1)$ -torus.

If L is flag, then $\pi_1(X_L) \cong A_L$ and X_L is aspherical.

FIX package cref breaks and some hyperlinks are wrong. Need to find an alternative.

8. CUBE COMPLEX

Definition 8.1. A cube complex is a space built out of Euclidean k -cubes $[0, 1]^k$ (k may be different) by gluing faces by (linear) isometries.

Example 8.2 (Examples of cube complex). • $\mathbb{S}^1$ and any 1-dimensional CW complex.

- Tori.
- $\mathbb{S}^2$

- Closed orientable genus 2 surface Σ_2 built out of 8 pentagons (glue 4 pentagons at each vertex). Note that by connecting the face center and five edge centers, we can subdivide a pentagon into 5 squares.
- Poincare homology sphere (See Example 6.9).
- Seifert–Weber dodecahedral space.

Definition 8.3. In a cube complex X , let v be a vertex. $lk(v)$ is the boundary of an ϵ -neighborhood of v in X (where *epsilon* is small, e.g. $1/3$). It consists of a collection of spherical simplices and is a Δ -complex.

Example 8.4.

- For a 1-simplex, $lk(v)$ is a point (that is *epsilon*-away from v).
- For a 2-simplex, $lk(v)$ is an arc (of a circle with radius *epsilon*).
- Glue the two adjacent edges of v together, then $lk(v)$ is a loop.

Definition 8.5 (Gromov’s Link Condition). A cube complex is *non-positively curved (NPC)* if for any vertex v , $lk(v)$ is simplicial and flag.

Definition 8.6. An NPC cube complex is CAT(0) if it’s simply-connected.

Note that this means any NPC cube complex admits a CAT(0) cube complex universal cover.

Theorem 8.7 (Gromov). *All CAT(0) cube complexes are contractible. Hence all connected NPC cube complexes are aspherical.*

Note that the converse is not true. Consider 3 squares glue to one vertex (a corner of the shell of a cube). This space is contractible, but the link is an empty triangle, hence this space is not NPC.

Observe that all graphs are NPC, as their links are disjoint points.

Proposition 8.8. *Let X be a 2-dimensional cube complex. Observe that all links are graph. The space is NPC if and only if each link is simplicial and has no loops of length 4.*

Lemma 8.9. *If L is a flag simplicial complex, then its Salvetti complex X_L (See Definition 7.7) is NPC.*

Note that the link of the only vertex in X_L may not be L .

Proof. A k -simplex in L gives a $k + 1$ -cube(torus) in X_L . The link then is joint of $k + 1$ $\mathbb{S}^0$ ’s, which is simplicial.

TODO Exercise: if L is flag then the link is also flag. □

Example 8.10 (Examples of fundamental groups of NPC cube complex). • Free groups.

- Free abelian groups.
- Fundamental groups of closed surfaces.
- (Khan-Markovic, Bergeron-Wise) If M is a closed hyperbolic 3-manifold, then there exists an NPC cube complex X such that $\pi_1(X) = \pi_1(M)$. This theorem is a key ingredient of Agol's Virtual Haken theorem and Virtual Fibration theorem.

An advantage of cube complex over simplicial complex is the hyperplanes.

Definition 8.11. TODO a formalized way to define hyperplane.

Hyperplanes are naturally immersed in the cube complex. For CAT(0) cube complex, the hyperplanes are embedded. Thus hyperplanes allows us to investigate some codimension-1 behavior of the spaces.

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